Maheen Hamid

AI Engineer | Generative AI · RAG · LLM Evaluation

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Professional Summary

Computer Science graduate focused on AI Engineering and Generative AI, with hands-on experience building Retrieval-Augmented Generation (RAG) systems, retrieval evaluation pipelines, vector search, LLM-based answer evaluation, and backend APIs. Interested in developing reliable, production-oriented AI systems with strong retrieval, evaluation, observability, and system design practices.

Technical Skills

Languages: Python, SQL, TypeScript
Generative AI: LLMs, RAG, Embeddings, Reranking, Prompt Engineering, Agentic Workflows
AI Frameworks: LangGraph, LangChain, Hugging Face
Machine Learning: PyTorch, scikit-learn, Model Evaluation, Fine-Tuning
Backend & APIs: FastAPI, REST APIs, Async Python
Databases & Retrieval: PostgreSQL, Qdrant, pgvector, Redis
DevOps: Docker, Git, GitHub Actions
AI Evaluation: RAG Evaluation, Faithfulness, Answer Relevance, Correctness, Hallucination Analysis

Professional Skills: Technical Communication, Problem Decomposition, Root-Cause Debugging, Requirements Translation, Technical Documentation, Cross-Functional Collaboration

Projects

Ragify – RAG Evaluation & Optimization Platform | GitHub — Oct. 2026 – Present
Python, FastAPI, React, TypeScript, PostgreSQL, Qdrant, Docker, Groq, Sentence Transformers

- Developed a full-stack RAG evaluation platform that benchmarks **Semantic, BM25, and Hybrid retrieval** configurations over user-uploaded document corpora using FastAPI, PostgreSQL, Qdrant, and React.
- Implemented automated synthetic evaluation datasets and retrieval metrics including **Hit@K, Precision@K, Recall@K, and Mean Reciprocal Rank (MRR)** to objectively compare RAG configurations.
- Integrated LLM-based answer evaluation through Groq to measure **faithfulness, relevance, correctness, and hallucination**, with deterministic failure classification, quality guardrails, and weighted recommendation scoring.
- Built project-scoped APIs, persistent experiment reporting, automated tests, and a responsive frontend that presents the recommended configuration, strengths, weaknesses, and actionable next steps.

Education

University of Sialkot — Sialkot, Pakistan
Bachelor of Science in Computer Science | CGPA: 3.77/4.00 — *Sep. 2022 – Sep. 2026*

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Natural Language Processing, Data Structures & Algorithms, Database Systems, Parallel & Distributed Computing

Awards & Achievements

Certificate of Appreciation – Department of Computer Science — 2026

- Awarded by the Department of Computer Science, University of Sialkot, for achieving a **4.00/4.00 GPA** in the 7th semester.